MIL-C-55021/2A
8 November 1973
SUPERSEDING
MIL-C-55021/2
10 December 1965

MILITARY SPECIFICATION

*CABLES: TWISTED PAIRS AND TRIPLES INTERNAL HOOKUP, SHIELDED AND UNSHIELDED, JACKETED OR UNJACKETED HIGH TEMPERATURE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

* 1.1 Scope. This specification covers polytetrafluoroethylene insulated electrical hookup wire in twisted pair and triples, shielded and unshielded, jacketed or unjacketed, for application in internal wiring of electrical and electronic equipment at temperatures from -40 degrees to 105 degrees C and from -65 degrees to 200 degrees C, with potential ratings from 600 to 1000 volts (rms) depending upon construction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-C-55021	Cables; Twisted Pairs and Triples, Internal Hookup, General Specification for

FSC 6145

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. The complete requirements for procuring cable described herein shall consist of this specification and the issue in effect of MIL-C-55021.

3.2 Material. All material used in construction of cables furnished under the specification shall meet the requirements specified herein.

3.2.1 Insulated conductors. Insulated conductors used in construction of cables furnished under this specification shall conform to the requirements of MIL-W-16878, type E or EE, as specified (see 6.2 of MIL-C-55021).

3.3 Construction

3.3.1 Type of wire. The type of wire shall be identified by the symbols "E" or "EE". All wire in the cable shall be of the same type and size (AWG).

3.3.2 Length of twist. The length of twist shall be as specified in table I.

Table I - Length of Twist

Wire Size (AWG)	Twist (Inches)	Twist Tolerance (Inches)
32 to 18	2.0 (50.8)	±0.5 (12.7)
16 to 14	2.5 (63.5)	±0.5
12 to 10	3.0 (76.2)	±0.5
8	4.0 (101.6)	±0.5

NOTE: Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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* 3.3.3 Shielding. When specified the shielding over the twisted cable shall conform to the requirements of MIL-W-16878 except that the minimum allowable coverage shall be 80 percent, and the angle made by the shielding braid with the longitudinal axis of the twisted conductors shall be between 20 and 40 degrees. The shield shall have good pushback characteristics.

* 3.3.4 Jacket. When specified the cable jacket shall be polyvinylchloride (PVC) extruded polytetrafluoroethylene (TFE) or perfluoropylene (FEP), or TFE tape wrap of uniform thickness as specified in Table II. If the jacket is tape, two layers of tape shall be applied in opposite directions, each layer and adjacent convolutions being properly sealed together to form a homogeneous wall. The surface of the jacket shall be smooth, although a slight spiral ridge of the tape on the jacket shall not be cause for rejection. Prefused tape shall not be used.

Table II - Jacket Thickness

Diameter Over Braided Shield	Jacket Thickness Nominal	Jacket Thickness Minimum
Inch	Inch	Inch
Under 0.300	.012	.008
0.300 and Over	.016	.012

* 3.4 Construction. The cable furnished under this specification shall be constructed in accordance with the requirements in tables I and II. The cable shall be capable of continuous operation in a temperature range of -62 degrees centigrade (C) (-80 degrees fahrenheit (F)) to 200 degrees C (392 degrees F) and shall be capable of continuous 600 volt root mean square (rms) operation.

NOTE: If PVC jacket is used over the construction, the temperature rating is -40°C to 105°C.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection provisions

4.1.1 Examination. Examination shall be in accordance with MIL-STD-105. The AQL shall be 4.0 percent. Visual examination shall be conducted for the purpose of determining compliance with the requirements for materials, construction, color, finish, product marking and workmanship.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-55021.

6. NOTES

6.1 Intended use. Cables covered by this specification are used in the hookup of various electronic assemblies. All cables are suitable for high frequency use. The shielded wires are used to provide a restriction of electrostatic voltages to the area of the wire and to restrict pickup, by the wire, of unwanted stray electrostatic charges.

6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Review Activities:
Army - EL
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